Optimization as the Engine of Machine Learning

Week 1 Lecture: objectives, derivatives, and optimality

A. D. Ignatov
HSE University, Faculty of Computer Science

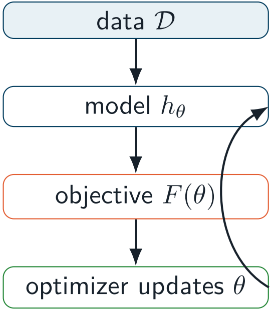
2026–2027

A model learns by solving an optimization problem

Given data $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^n$, choose parameters θ by

$$\min_{\theta \in \Theta} F(\theta) = \frac{1}{n} \sum_{i=1}^{n} \ell(h_\theta(x_i), y_i) + \lambda R(\theta).$$

- h_θ: model and its parameters;
- ℓ: how wrong a prediction is;
- R: preference for simpler solutions;
- Θ: constraints on valid parameters.



By the end of today, you should be able to

1. translate a learning problem into variables, objective, and constraints;
2. compute gradients and Hessians using differentials;
3. distinguish local, global, strict minima, and saddle points;
4. prove first- and second-order optimality conditions;
5. derive the objectives and gradients of linear and logistic regression;
6. check an analytic gradient numerically before trusting it.

Takeaway. Optimization is not merely a solver call: the objective encodes what the model will learn.

Every optimization problem has three ingredients

$$\boxed{\min_{x \in \mathcal{X}} f(x)}$$

Decision variable x

- model weights;
- allocation vector;
- schedule or route.

Objective f

- prediction error;
- cost or risk;
- negative reward.

Feasible set $\mathcal{X}$

- $\mathbb{R}^d$;
- norm or box bounds;
- structural constraints.

Maximization is equivalent to minimizing $-f$. Vector variables can contain every trainable parameter of a model.

The loss function defines the behavior we reward

Model	Per-example loss	Typical objective
Linear regression	$\frac{1}{2}(\hat{y} - y)^2$	mean squared error
Logistic regression	$\log(1 + e^{-yz})$	negative log-likelihood
Multiclass classifier	$-\log p_y$	cross-entropy
Matrix factorization	$(a_{ij} - u_i^\top v_j)^2$	observed reconstruction error
Support vector machine	$\max(0, 1 - yz)$	hinge loss + regularization

Question. If two objectives fit the training data equally well, what should decide between their solutions?

Local and global minima answer different questions

Local minimum

$x^\star \in \mathcal{X}$ is local if some $r > 0$ satisfies

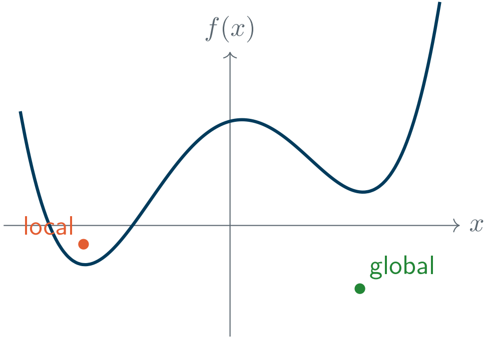
$$f(x^\star) \leq f(x) \quad \forall x \in \mathcal{X} \cap B_r(x^\star).$$

Global minimum

$x^\star$ is global if

$$f(x^\star) \leq f(x) \quad \forall x \in \mathcal{X}.$$

Strict minima replace $\leq$ by $<$ for $x \neq x^\star$.



Existence and uniqueness are separate issues

A minimum may not exist

$$\min_{x \in \mathbb{R}} e^x$$

The infimum is 0, but no finite x attains it.

A minimum may not be unique

$$\min_{x \in \mathbb{R}} (x^2 - 1)^2$$

Both $x = -1$ and $x = 1$ are global minimizers.

Weierstrass theorem

If $\mathcal{X}$ is nonempty and compact and f is continuous on $\mathcal{X}$, then f attains a global minimum and maximum.

- Existence: often obtained from compactness or coercivity.
- Uniqueness: often obtained from strict convexity.

The derivative is the best local linear model

For differentiable $f : \mathbb{R}^d \to \mathbb{R}$,

$$f(x+h) = f(x) + \langle \nabla f(x), h \rangle + o(\|h\|).$$

The vector $\nabla f(x)$ is defined by the linear term in every direction h.

Directional derivative

$$D_d f(x) = \lim_{t \downarrow 0} \frac{f(x+td) - f(x)}{t} = \langle \nabla f(x), d \rangle .$$

By Cauchy–Schwarz,

$$\langle \nabla f, d \rangle \geq - \|\nabla f\| \, \|d\| .$$

For $\|d\| = 1$, steepest local decrease uses

$$d = -\frac{\nabla f(x)}{\|\nabla f(x)\|}.$$

Why every interior minimum is stationary

Theorem

Let $f : \mathbb{R}^d \to \mathbb{R}$ be differentiable near an unconstrained local minimizer $x^\star$. Then

$$\nabla f(x^\star) = 0.$$

Proof. Fix any direction d and define $\phi(t) = f(x^\star + td)$. Since $x^\star$ is a local minimizer of f, $t = 0$ is a local minimizer of ϕ. Therefore

$$0 = \phi'(0) = \langle \nabla f(x^\star), d \rangle .$$

This holds for every d. Choosing $d = \nabla f(x^\star)$ gives $\|\nabla f(x^\star)\|^2 = 0$. Hence $\nabla f(x^\star) = 0$. $\square$

Takeaway. Stationarity is necessary, not sufficient: maxima and saddle points are stationary too.

Constraints change the first-order condition

For $\min_{x \in \mathcal{X}} f(x)$, a vector d is *feasible at* x if $x + td \in \mathcal{X}$ for all sufficiently small $t \geq 0$.

First-order necessary condition

If $x^\star$ is a local minimum and f is differentiable, then

$$\langle \nabla f(x^\star), d \rangle \geq 0 \quad \text{for every feasible direction } d.$$

Proof. For a feasible d, the function $\phi(t) = f(x^\star + td)$ has a one-sided minimum at $t = 0$. Thus $\phi'(0^+) \geq 0$, and the chain rule gives the claim. $\square$ At an interior point, both d and $-d$ are feasible, so the condition reduces to $\nabla f(x^\star) = 0$.

The Hessian describes local curvature

If f is twice continuously differentiable,

$$\nabla^2 f(x) = H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_d} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_d \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_d^2} \end{bmatrix}.$$

Clairaut's theorem gives $H_f(x) = H_f(x)^\top$ when mixed derivatives are continuous.

- $d^\top H_f(x) d > 0$: positive curvature along d;
- $d^\top H_f(x) d < 0$: negative curvature along d;
- eigenvectors identify principal curvature directions.

Definiteness is equivalent to the signs of eigenvalues

Theorem

For a symmetric matrix H, $H \succeq 0$ if and only if every eigenvalue of H is nonnegative. Likewise, $H \succ 0$ iff every eigenvalue is positive.

Proof. By the spectral theorem, $H = Q\Lambda Q^\top$ with orthogonal Q. For $z = Q^\top x$,

$$x^\top Hx = z^\top \Lambda z = \sum_{i=1}^{d} \lambda_i z_i^2.$$

If all $\lambda_i \geq 0$, the sum is nonnegative. Conversely, for an eigenvector q_i, $q_i^\top Hq_i = \lambda_i \|q_i\|^2$; positive semidefiniteness forces $\lambda_i \geq 0$. The strict case is analogous. □

Taylor's theorem turns curvature into optimality tests

For $f \in C^2$ near x,

$$f(x+h) = f(x) + \nabla f(x)^\top h + \frac{1}{2} h^\top H_f(x) h + o(\|h\|^2).$$

At a stationary point, the quadratic term becomes decisive:

Hessian at $x^\star$	Local geometry	Conclusion
$H \succ 0$	curves upward in every direction	strict local minimum
$H \prec 0$	curves downward in every direction	strict local maximum
H indefinite	both upward and downward directions	saddle point
$H \succeq 0$ but singular	flat directions exist	inconclusive

A minimum cannot curve downward

Theorem

If $x^\star$ is an unconstrained local minimizer and f is twice differentiable, then

$$\nabla f(x^\star) = 0, \qquad H_f(x^\star) \succeq 0.$$

Proof. First-order necessity gives $\nabla f(x^\star) = 0$. For any d, Taylor expansion along td yields

$$f(x^\star + td) - f(x^\star) = \frac{t^2}{2} d^\top H_f(x^\star) d + o(t^2).$$

The left side is nonnegative for small t. Divide by $t^2/2$ and let $t \to 0$: $d^\top H_f(x^\star) d \geq 0$ for every d. Hence $H_f(x^\star) \succeq 0$. $\square$

Positive curvature certifies a strict minimum

Theorem

If $\nabla f(x^\star) = 0$ and $H_f(x^\star) \succ 0$, then $x^\star$ is a strict local minimizer.

Proof. Let $m = \lambda_{\min}(H_f(x^\star)) > 0$. Continuity of the Hessian implies that in a small neighborhood, $h^\top H_f(x)h \geq \frac{m}{2} \|h\|^2$. Taylor's theorem in mean-value form gives, for some point ξ on the segment from $x^\star$ to $x^\star + h$,

$$f(x^\star + h) - f(x^\star) = \frac{1}{2} h^\top H_f(\xi) h \geq \frac{m}{4} \|h\|^2 > 0$$

for every sufficiently small nonzero h. □ Why not merely $H \succeq 0$? For $f(x) = x^3$, both $f'(0)$ and $f''(0)$ vanish, but 0 is not a minimum.

Three stationary points, three different outcomes

Strict minimum

$$f(x,y) = x^2 + 2y^2$$

$$H = \mathrm{diag}(2,4) \succ 0.$$

Strict maximum

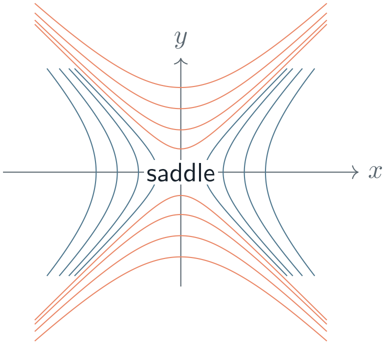
$$f(x,y) = -x^2 - y^2$$

$$H = -2I \prec 0.$$

Saddle

$$f(x,y) = x^2 - y^2$$

$$H = \mathrm{diag}(2,-2).$$



High-dimensional losses have saddle directions

At a stationary point, the Hessian may have positive, negative, and nearly zero eigenvalues.

- A negative eigenvalue proves that a descent direction exists locally.
- Near-zero eigenvalues create flat valleys and slow numerical progress.
- In neural networks, parameter symmetries often create entire families of equivalent solutions.
- A low training objective does not imply good test performance.

Takeaway. The optimizer sees local geometry; the learning problem also cares about data, regularization, and generalization.

Differentials make matrix derivatives systematic

For scalar f, identify the coefficient of dx in

$$df = \langle \nabla f(x), dx \rangle .$$

Useful rules:

$$d(Ax) = A\,dx, \qquad d(x^\top y) = y^\top dx + x^\top dy,$$
$$d(x^\top Ax) = x^\top A\,dx + x^\top A^\top dx = \left\langle (A + A^\top)x, dx \right\rangle .$$

Therefore

$$\nabla_x (x^\top Ax) = (A + A^\top)x,$$

which becomes $2Ax$ when A is symmetric.

A quadratic reveals its geometry exactly

Consider

$$f(x) = \frac{1}{2}x^\top Ax - b^\top x + c, \qquad A = A^\top.$$

Then

$$df = \langle Ax - b, dx \rangle, \qquad \nabla f(x) = Ax - b, \qquad H_f(x) = A.$$

If $A \succ 0$, the unique stationary point is

$$x^\star = A^{-1}b.$$

Moreover,

$$f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top A(x - x^\star) > 0$$

for $x \neq x^\star$, proving global optimality and uniqueness.

Linear regression is a quadratic optimization problem

Let $X \in \mathbb{R}^{n\times d}$, $y \in \mathbb{R}^n$, and predictions $\hat{y} = Xw$:

$$F(w) = \frac{1}{2n} \|Xw - y\|_2^2 .$$

Using $r = Xw - y$,

$$dF = \frac{1}{n} r^\top dr = \frac{1}{n} r^\top X\, dw = \left\langle \frac{1}{n} X^\top (Xw - y), dw \right\rangle .$$

Hence

$$\boxed{\nabla F(w) = \frac{1}{n} X^\top (Xw - y)}, \qquad \boxed{H_F(w) = \frac{1}{n} X^\top X \succeq 0}.$$

The Hessian is positive definite exactly when X has full column rank.

Full rank makes least squares unique

Stationarity gives

$$X^\top X w = X^\top y.$$

If X has full column rank, then $X^\top X \succ 0$ and

$$w^\star = (X^\top X)^{-1} X^\top y.$$

Proof of positive definiteness. For $v \neq 0$,

$$v^\top X^\top X v = \|Xv\|_2^2 > 0$$

because full column rank implies $Xv \neq 0$. Therefore the objective is strictly convex and $w^\star$ is the unique global minimizer. $\square$ Numerical note: solve the linear system; do not explicitly form an inverse in production code.

Ridge regularization repairs non-uniqueness

$$F_\lambda(w) = \frac{1}{2n}\,\|Xw - y\|_2^2 + \frac{\lambda}{2}\,\|w\|_2^2\,, \qquad \lambda > 0.$$

Then

$$\nabla F_\lambda(w) = \frac{1}{n}X^\top(Xw - y) + \lambda w,$$

$$H_{F_\lambda}(w) = \frac{1}{n}X^\top X + \lambda I \succ 0.$$

Thus the unique minimizer exists even when X is rank deficient:

$$w_\lambda^\star = \left(\frac{1}{n}X^\top X + \lambda I\right)^{-1}\frac{1}{n}X^\top y.$$

Takeaway. Regularization changes both statistical preference and optimization geometry.

Logistic regression has a smooth objective

For labels $y_i \in \{-1, +1\}$ and score $z_i = w^\top x_i$,

$$\ell_i(w) = \log\left(1 + e^{-y_i w^\top x_i}\right).$$

Let $s_i = -y_i w^\top x_i$. Since $d \log(1 + e^s)/ds = \sigma(s)$,

$$\nabla \ell_i(w) = -y_i\, \sigma(-y_i w^\top x_i) x_i.$$

Therefore

$$\nabla F(w) = \frac{1}{n} \sum_{i=1}^{n} -y_i \sigma(-y_i w^\top x_i) x_i + \lambda w.$$

Its Hessian is a weighted sum of $x_i x_i^\top$ plus λI, hence positive semidefinite for $\lambda \geq 0$.

Stable formulas matter as much as correct calculus

The expression $\log(1+e^z)$ overflows for large positive z. Use the identity

$$\log(1+e^z) = \max(0,z) + \log(1+e^{-|z|}).$$

Likewise, avoid explicitly computing matrix inverses and avoid subtracting nearly equal numbers.

Mathematically correct

- exact formula;
- exact derivative;
- exact stationary equation.

Numerically reliable

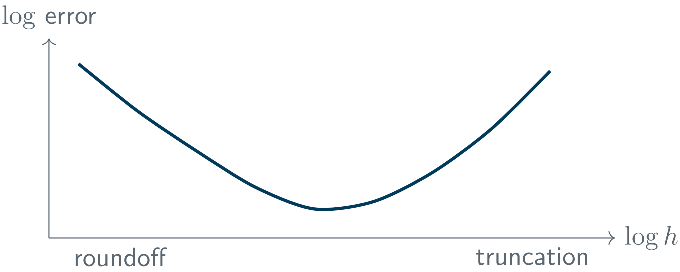
- stable evaluation;
- controlled conditioning;
- tested implementation.

Finite differences diagnose gradient errors

For coordinate vector e_j,

$$\frac{\partial f(x)}{\partial x_j} \approx \frac{f(x + he_j) - f(x - he_j)}{2h}.$$

Central differences have truncation error $O(h^2)$, but roundoff grows when h is too small.



Use relative error $\|g_{\text{analytic}} - g_{\text{FD}}\| / (1 + \|g_{\text{analytic}}\| + \|g_{\text{FD}}\|)$.

Derivative evaluation determines the cost

For dense linear regression with $X \in \mathbb{R}^{n \times d}$:

- objective evaluation: $O(nd)$;
- gradient evaluation: $O(nd)$;
- explicit Hessian $X^\top X$: $O(nd^2)$;
- storing the Hessian: $O(d^2)$;
- finite-difference gradient: roughly $2d$ objective calls.

Matrix association matters: compute $X^\top(Xw - y)$, not $(X^\top X)w - X^\top y$, when n and d make forming $X^\top X$ expensive or ill-conditioned.

Takeaway. An algorithm is a mathematical rule plus a computational strategy.

A practical checklist for a new learning objective

1. Define parameters, data, objective, regularizer, and constraints.
2. Confirm shapes and scaling: sum or mean? per sample or per batch?
3. Derive the differential, gradient, and—when useful—the Hessian.
4. Check stationary points and curvature on a small instance.
5. Implement a numerically stable objective.
6. Compare the analytic gradient with finite differences.
7. Only then select and tune an optimization algorithm.

What we proved—and what remains

- A differentiable unconstrained local minimum must be stationary.
- Its Hessian must be positive semidefinite.
- A stationary point with positive definite Hessian is a strict local minimum.
- Linear regression is convex; full rank or ridge regularization makes it strictly convex.
- Correct derivatives need stable and efficient implementations.

Next week: convexity will turn local information into global guarantees, and KKT conditions will organize constrained optimality.

Sources and further reading

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